

Studio di funzione - Esercizi

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ESERCIZIO 1

$$Y = 2x^3 + x$$

1. DOMINIO $D = \mathbb{R}$ $D = (-\infty, +\infty)$

2. SEGNO

$$f(-x) = f(x) \quad \text{NON È PARI}$$

$$\begin{aligned} f(-x) &= -f(x) & 2(-x)^3 + (-x) &= -(2x^3 + x) \\ & & -2x^3 - x &= -2x^3 - x \quad \text{È} \end{aligned}$$

3. INTERSEZIONI

$$\begin{cases} 2x^3 + x = y \\ x = 0 \end{cases} \rightarrow \begin{cases} y = 0 \\ x = 0 \end{cases} \quad (0, 0)$$

$$\begin{cases} 2x^3 + x = 0 \\ y = 0 \end{cases} \rightarrow \begin{cases} x(2x^2 + 1) = 0 \\ y = 0 \end{cases} \quad \begin{array}{l} F_1 \quad x = 0 \\ F_2 \quad 2x^2 = -1 \\ (0, 0) \end{array}$$

4. SEGNO

$$2x^3 + x > 0 \rightarrow x(2x^2 + 1) > 0$$

$$F_1 \quad x > 0$$

$$F_2 \quad 2x^2 + 1 > 0 \rightarrow 2x^2 > -1 \rightarrow x^2 > -\frac{1}{2} \quad \forall x$$

$$\begin{array}{c} F_1 \\ F_2 \end{array} \quad \begin{array}{c} - & 0 & + \\ + & | & + \\ - & | & + \end{array}$$

$$I_p = (0, +\infty)$$

$$I_m = (-\infty, 0)$$

$$y = \frac{x^2 - 1}{x}$$

INTERSECTION 1

$$\begin{cases} y = \frac{x^2 - 1}{x} \\ x = 0 \end{cases} \rightarrow \begin{cases} y = \frac{-1}{0} \\ x = 0 \end{cases} \quad \text{impossible}$$

$$\begin{cases} \frac{x^2 - 1}{x} = 0 \\ y = 0 \end{cases} \rightarrow \begin{cases} x = \pm 1 \\ y = 0 \end{cases} \quad \begin{matrix} (-1, 0) \\ (1, 0) \end{matrix}$$